

Formal Language Bias in LLMs

Measuring and Mitigating Formal Register Bias in Large Language

Model Outputs

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Abstract

Large language models are increasingly deployed as conversational agents, yet their responses often sound more formal than ordinary human dialogue. This project investigates whether that tendency can be measured as a systematic formality bias rather than treated as an anecdotal stylistic observation. We define formal language bias as a tendency for LLM outputs to prefer elevated, document-like, or institutionally formal register even in contexts where more casual human language would be expected. To measure this behavior, we construct a dual formality framework consisting of a lexicon-based Formality Rate metric and a RoBERTa-based register classifier trained on the Pavlick formality scores dataset. We evaluate LLM outputs from LMSYS-Chat-1M against multiple human baselines, including Reddit, ELI5, and BlendedSkillTalk, and analyze variation across conversation type and model family. We further test whether lightweight LoRA fine-tuning on low-formality data can reduce formal register bias in Qwen2.5-3B-Instruct generations. Our results show that LLM responses exhibit substantially higher Formality Rate and RoBERTa formal probability than human conversational baselines. The effect is widespread across model families but varies by model family and conversation type. LoRA mitigation produced lower mean lexical formality and neural formality scores in a small held-out evaluation, suggesting that register behavior may be partially steerable through fine-tuning. However, the results also show that formality is highly dependent on dataset genre, metric design, model family, and task context.

1. Introduction

Large language models have become central to modern natural language interfaces. They are used for search assistance, tutoring, writing support, summarization, customer service, coding help, and open-ended conversation. As these systems become more conversational, their quality is no longer determined only by factual correctness, grammaticality, or coherence. A useful conversational model must also produce language that fits the social and pragmatic context of the interaction. In other words, it must choose an appropriate register.

Register refers to a style of language associated with a particular context, audience, or communicative purpose. A formal register may be appropriate in an academic report, legal explanation, or professional memo. However, in casual conversation, overly formal language can feel unnatural, distant, or misaligned with the user. For example, an assistant might prefer words such as “utilize,” “commence,” “facilitate,” or “delve” when a human speaker would more naturally write “use,” “start,” “help,” or “dig into.” This is not necessarily a semantic error. The

meaning may be correct, but the language style may not match the expectations of everyday conversation.

This project studies that phenomenon as **formal language bias**. We define formal language bias as a systematic tendency for LLM outputs to favor formal-register language above the level observed in comparable human conversational corpora. This bias matters because language style affects how users perceive accessibility, naturalness, trust, and alignment. A response that is factually correct but stylistically distant may be less effective in contexts such as education, mental-health support, consumer assistance, peer tutoring, and informal dialogue.

The project is motivated by a common observation sometimes described as the “delve problem”: LLMs appear to overuse polished, elevated, or document-like phrasing. The goal of this work is to move beyond anecdotal examples and evaluate whether this stylistic tendency can be measured empirically. To do this, we compare LLM outputs with multiple human baselines and evaluate formality through two complementary methods. The first is a transparent lexicon-based metric called Formality Rate, which measures formal lexical markers per 1,000 tokens. The second is a neural register classifier based on RoBERTa, which estimates whether a response is likely to belong to a formal register.

The central claim of this report is that LLM formality bias is measurable, widespread, and partially steerable, but also difficult to interpret without careful attention to baseline selection, topic effects, and metric limitations.

2. Research Questions and Contributions

2.1 Research Questions

This project is organized around four research questions.

RQ1: Do LLM outputs exhibit higher formal register than human conversational corpora?

The first goal is to determine whether LLM responses are measurably more formal than human-written conversational baselines. This requires comparing LLM outputs not only to one human dataset, but to several human corpora with different genres and levels of expected formality.

RQ2: Can lexical and neural metrics both detect formality differences across corpora and model families?

A purely lexical metric may miss contextual style, while a neural classifier may be less interpretable. We therefore ask whether both approaches support the same broad conclusion.

RQ3: Does formal language bias vary by conversation type and model family?

Formality may depend on whether a response is casual, task-oriented, or general. It may also vary across GPT, LLaMA, Claude, PaLM, Alpaca, Vicuna, and other model families. This question examines whether the bias is universal or shaped by training and alignment differences.

RQ4: Can lightweight fine-tuning reduce formal register bias?

Finally, we test whether LoRA fine-tuning on low-formality data can shift a model’s register downward while preserving reasonable response quality.

2.2 Contributions

This project makes four main contributions.

First, we construct a dual formality measurement framework. The first metric, Formality Rate, directly measures the frequency of formal lexical markers per 1,000 tokens. The second metric, a RoBERTa register classifier, estimates the probability that a response belongs to a formal register. These two metrics capture different dimensions of style: surface lexical choice and broader contextual formality.

Second, we compare LLM outputs against multiple human baselines rather than using a single reference corpus. This is important because “human language” is not a uniform category. Reddit and BlendedSkillTalk represent more casual and conversational language, while ELI5 represents explanatory human writing that is naturally more formal. This allows us to distinguish general human informality from genre-specific explanatory formality.

Third, we conduct a cross-model and cross-context analysis. By grouping responses by model family and conversation type, we evaluate whether formality bias is isolated to a specific model or appears across different families and interaction contexts.

Fourth, we test a proof-of-concept mitigation strategy using LoRA fine-tuning on Qwen2.5-3B-Instruct. This experiment evaluates whether formal register behavior can be shifted through lightweight adaptation without updating all model parameters.

3. Related Work

3.1 Formality Detection

Prior work on formality detection has shown that formality can be identified using lexical, syntactic, and statistical features. Pavlick and Tetreault established important foundations for automatic formality detection by showing that formality can be predicted from surface-level and linguistic features. This motivates our use of a lexicon-based formality metric. Formality Rate is intentionally simple and interpretable: it counts formal words and phrases directly. This makes it easy to inspect which lexical choices drive the score.

However, formality is not only a matter of individual words. A response can sound formal because of sentence structure, discourse organization, hedging, or explanatory framing. For this reason, we also use a neural classifier to capture stylistic cues that a lexicon may miss.

3.2 Formality Scoring and the Pavlick Dataset

Pavlick and Tetreault’s formality scoring work provides the supervised foundation for our RoBERTa-based register classifier. Rather than training on formal/informal rewrite pairs, our classifier uses the Pavlick formality scores dataset, which contains text examples associated with formality labels or scores. This makes the classifier better aligned with our goal: estimating register formality rather than performing style transfer.

This distinction matters because our project is not primarily a text-rewriting system. We are not asking the model to convert formal sentences into informal ones. Instead, we use supervised formality data to train a classifier that can assign formality probabilities to unseen LLM and human responses. These probabilities are then used as a neural complement to the lexicon-based Formality Rate metric.

3.3 Human-Bot Register Differences

Recent work comparing human-human and human-bot conversations suggests that conversational agents exhibit systematic stylistic and register differences. This project builds on that idea by focusing specifically on formality as one measurable dimension of register. Instead of only asking whether humans and bots differ in general, we ask whether LLMs consistently produce more formal language than human conversational corpora.

3.4 Pretraining, Alignment, and Linguistic Bias

Prior research on pretraining and linguistic bias has shown that model behavior reflects the data and optimization procedures used during training. This motivates our hypothesis that formal language bias may arise from a combination of pretraining corpora, supervised instruction tuning, RLHF, and product-oriented alignment. Large-scale web text contains many formal genres, including articles, documentation, academic content, manuals, and institutional writing. Instruction tuning may further reinforce polished assistant-like responses. As a result, LLM formality may not come from one isolated source, but from the interaction between data composition and alignment objectives.

4. Data

4.1 LMSYS-Chat-1M

The main LLM corpus for this project is LMSYS-Chat-1M. This dataset contains large-scale user-assistant conversations involving many different model families. It is useful for this project because it provides real conversational LLM outputs rather than isolated benchmark completions. It also allows comparisons across model families and conversation types.

For each LMSYS example, we isolate assistant responses and treat those responses as the primary LLM output corpus. The analysis focuses on the language generated by the model, not the user prompt. Where possible, model names are parsed and grouped into families such as GPT, LLaMA, Claude, PaLM, Alpaca, and Vicuna. This grouping enables comparison of formality patterns across model families.

4.2 Human Baseline Corpora

A central challenge in this project is choosing an appropriate human baseline. Human text varies widely by genre, so no single dataset can represent all human conversation. To address this, we compare LLM outputs against three different human baselines.

Reddit is used as an informal web-text baseline. It represents casual online writing and is expected to have relatively low formal register. Reddit is useful because it provides a strong contrast against polished assistant-style responses.

BlendedSkillTalk is used as a dialogue-oriented human baseline. Because it consists of scripted or dialogue-like interactions, it better approximates conversational exchange than explanatory web text. It is expected to be one of the least formal baselines.

ELI5 is used as an explanatory human baseline. Although it is human-written, it is not purely casual. ELI5 responses often explain concepts in a structured way, which naturally increases formality. This makes ELI5 an important comparison point because many LLM responses are explanatory. If LLMs exceed even ELI5 in formality, that provides stronger evidence of formal register bias.

The use of multiple baselines is necessary because the interpretation of “formal bias” depends on the reference point. Comparing LLMs only to Reddit might exaggerate the effect, while comparing only to ELI5 might hide differences in casual contexts. A multi-baseline design allows a more nuanced analysis.

4.3 Pavlick Formality Scores Dataset

The Pavlick formality scores dataset is used for training the RoBERTa register classifier. This dataset provides supervised examples for learning formality as a register property. The classifier trained on this data is then used to assign a formal probability score to responses from LMSYS and the human baseline corpora. In this project, GYAFC is discussed as related work on formality style transfer, but it is not used as the classifier training source in the final code.

4.4 Preprocessing

Preprocessing is necessary because raw conversational and web-text data contain formatting artifacts that can distort formality measurements. Markdown, bullet lists, code blocks, copied prompt text, URLs, and malformed outputs can introduce tokens that do not reflect natural conversational register.

The preprocessing pipeline includes the following steps:

Step	Purpose
<i>Assistant-turn filtering</i>	Ensures that only model responses are scored in LMSYS
<i>Markdown and formatting cleanup</i>	Removes artifacts that may inflate formality or token counts
<i>Whitespace normalization</i>	Standardizes text before tokenization
<i>Lowercasing</i>	Supports consistent lexicon matching
<i>Regex tokenization</i>	Produces stable token counts without relying on external tokenizer state
<i>Length filtering</i>	Removes empty, extremely short, or malformed responses

<i>Source labeling</i>	Preserves corpus and model-family labels for aggregation
<i>Phrase matching before unigram matching</i>	Prevents undercounting multiword formal expressions

These preprocessing decisions directly affect both lexical and neural measurements. For example, markdown-heavy responses may appear more formal because they contain structured writing patterns rather than natural dialogue. Similarly, extremely short responses may produce unstable Formality Rate values because one formal word can dominate the normalized score.

5. Methodology

5.1 Formality Rate

The first metric is **Formality Rate**, a lexicon-based measure of formal lexical density. For a response x , Formality Rate is defined as:

$$FR(x) = 1000 \times \frac{formal_hits(x)}{token_count(x)}$$

where $formal_hits(x)$ is the number of formal words or phrases matched in the response, and $token_count(x)$ is the total number of tokens in the response. Multiplying by 1,000 normalizes scores across different response lengths.

The lexicon is organized into two tiers.

Tier 1: LLM-isms.

This tier contains words and phrases commonly associated with polished assistant language. Examples include words such as “delve,” “utilize,” “commence,” “facilitate,” “robust,” “comprehensive,” “underscore,” and “leverage.”

Tier 2: Broad formal vocabulary.

This tier contains more general formal-register markers that are not exclusive to LLMs but are associated with elevated or academic style. Examples include “therefore,” “moreover,” “furthermore,” “consequently,” “substantial,” “demonstrate,” and “significant.”

The advantage of Formality Rate is interpretability. If a corpus scores highly, we can inspect which words caused the score to increase. This makes FR useful for diagnosing surface-level register patterns. However, the metric also has limitations. It cannot detect formality caused by syntax, discourse structure, hedging, or explanatory framing unless those

features appear through explicit lexicon entries. It may also miss informal words used in otherwise formal sentences or formal words used casually.

5.2 RoBERTa Register Classifier

The second metric is a RoBERTa-based register classifier. Unlike Formality Rate, which counts predefined lexical markers, RoBERTa can capture contextual cues that extend beyond individual words.

The classifier uses a pretrained RoBERTa encoder followed by a binary classification head. Each input response is tokenized into subword units and passed through the RoBERTa encoder. The classification head maps the contextual representation to formal and informal logits. A softmax layer converts these logits into a probability distribution. We use the formal class probability as the neural formality score:

$$P_{formal}(x) = softmax(f_{\theta}(x))_{formal}$$

RoBERTa was selected because formality is often contextual. A response may sound formal because it uses longer sentence structures, explanatory framing, passive constructions, or discourse markers even when it does not contain obvious formal vocabulary. A transformer encoder is better suited to capturing these broader patterns than a lexicon-only method.

The classifier is trained on the Pavlick formality scores dataset and evaluated using standard classification metrics, including accuracy, precision, recall, and F1 score. Once trained, it is applied to LMSYS and human baseline responses to estimate formal-register probability across corpora.

5.3 Conversation-Type Analysis

Because formality depends heavily on communicative context, we group LMSYS responses by conversation type. The major categories are:

Conversation Type	Description
Casual	Informal or social dialogue
General	Broad informational or open-ended conversation
Task	Instructional, problem-solving, or goal-directed responses

This analysis is important because task-oriented responses are expected to be more structured and formal than casual ones. If LLMs are only formal in task-oriented settings, the effect may be

explained by topic or function. However, if casual LLM conversations remain more formal than human conversational baselines, that provides stronger evidence of register mismatch.

5.4 Model-Family Analysis

We also analyze formality by model family. Model-family grouping allows us to test whether formal language bias is concentrated in one type of model or appears across different architectures and alignment pipelines.

The analysis compares mean Formality Rate and mean RoBERTa P(formal) across families such as GPT, PaLM, LLaMA, Claude, Alpaca, and Vicuna. These comparisons are not causal, because model families differ in many ways, including pretraining data, instruction tuning, safety tuning, RLHF, system prompts, and deployment context. However, cross-family differences can suggest whether training and alignment choices may influence register behavior.

5.5 LoRA Mitigation

The third methodological component is a proof-of-concept mitigation experiment using LoRA fine-tuning. The goal is to test whether formal register behavior can be shifted downward through lightweight adaptation.

The base model for the updated mitigation experiment is Qwen2.5-3B-Instruct. We use LoRA to fine-tune the model on lower-formality examples while keeping the original model weights frozen. This allows us to test whether register can be shifted through lightweight adaptation without fully retraining the model. The experiment remains proof-of-concept because it uses a limited held-out prompt set, but it is stronger than the earlier small-model setup because the base model is an instruction-tuned chat model rather than GPT-2-medium.

LoRA, or Low-Rank Adaptation, freezes the original model weights and trains small adapter matrices inside selected transformer modules. This reduces the number of trainable parameters and helps avoid fully overwriting the model’s pretrained language knowledge. The model is fine-tuned on low-formality data and then evaluated on held-out prompts before and after adaptation.

Mitigation is evaluated using three main quantities:

$$\Delta FR = FR_{after} - FR_{before}$$

$$\Delta P_{formal} = P_{formal, after} - P_{formal, before}$$

and a fluency or similarity metric such as ROUGE-L, when applicable. Negative values for ΔFR and ΔP_{formal} indicate reduced formality after fine-tuning.

6. Experimental Setup

6.1 Experiment 1: LLMs vs Human Baselines

The first experiment compares LMSYS LLM responses against Reddit, BlendedSkillTalk, and ELI5. Each response is scored using Formality Rate and RoBERTa P(formal). We report mean scores, standard deviations, and distribution-level comparisons.

The main purpose of this experiment is to answer whether LLM outputs are more formal than human conversational corpora. Because Formality Rate distributions are sparse, skewed, and often zero-inflated, nonparametric testing is preferred. The Mann-Whitney U test is used to compare LLM and human distributions without assuming normality.

6.2 Experiment 2: Conversation-Type and Model-Family Analysis

The second experiment evaluates whether formality varies by conversation type and model family. LMSYS responses are grouped into casual, general, and task-oriented categories. Scores are then aggregated within each group.

Model-family analysis is conducted by parsing available model identifiers and grouping them into families. Mean Formality Rate and mean RoBERTa P(formal) are then compared across families. A Kruskal-Wallis test can be used when comparing more than two groups because the distributions may violate normality assumptions.

6.3 Experiment 3: LoRA Mitigation

The third experiment evaluates the effect of LoRA fine-tuning. A set of held-out LMSYS prompts is used to generate responses before and after LoRA adaptation. The same prompts are used in both conditions, which allows paired comparison.

For each prompt, we calculate pre-mitigation and post-mitigation Formality Rate and RoBERTa P(formal). We then compute the difference between the two conditions. A paired statistical test is appropriate because each post-fine-tuning response corresponds to the same prompt as a pre-fine-tuning response.

The mitigation experiment is evaluated using:

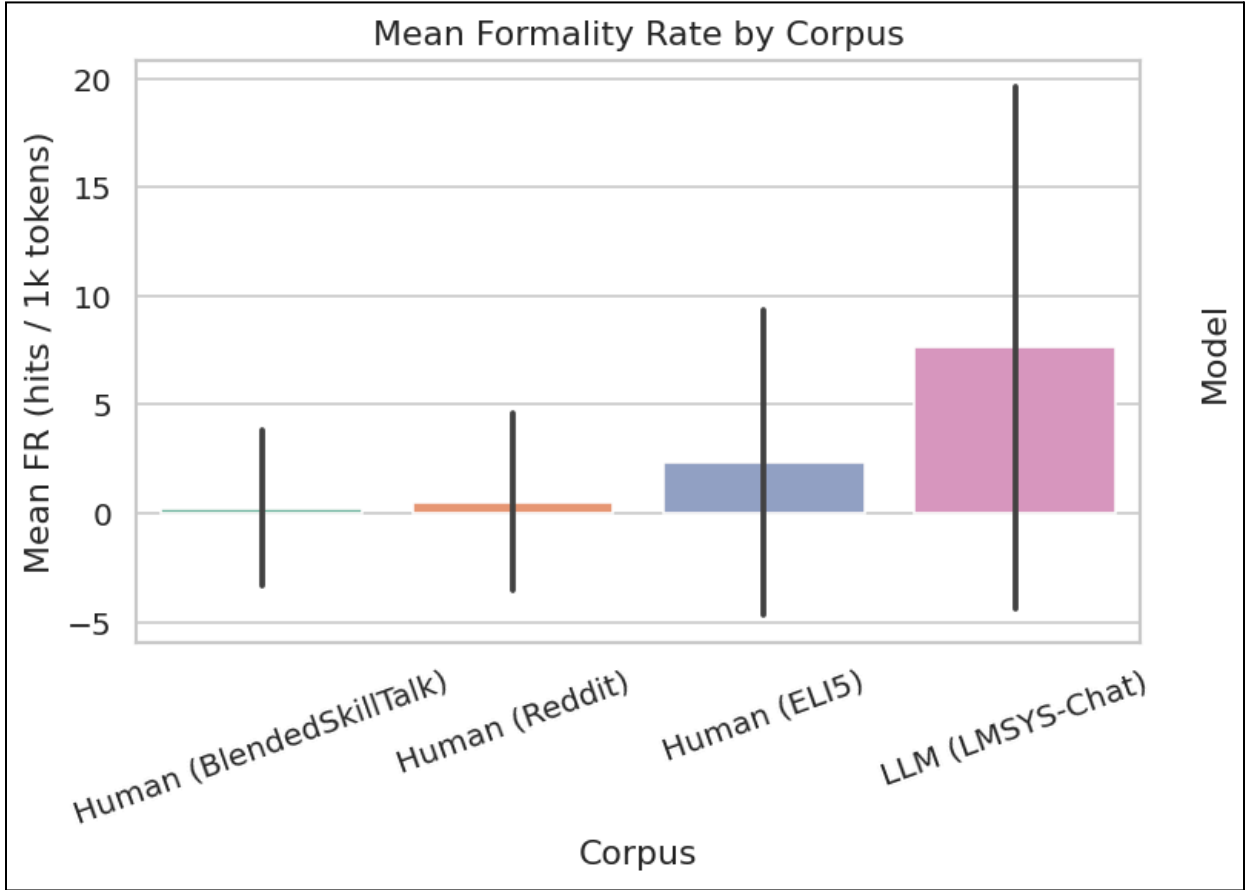
Metric	Purpose
Δ FR	Measures change in lexical formality
Δ RoBERTa P(formal)	Measures change in neural register classification

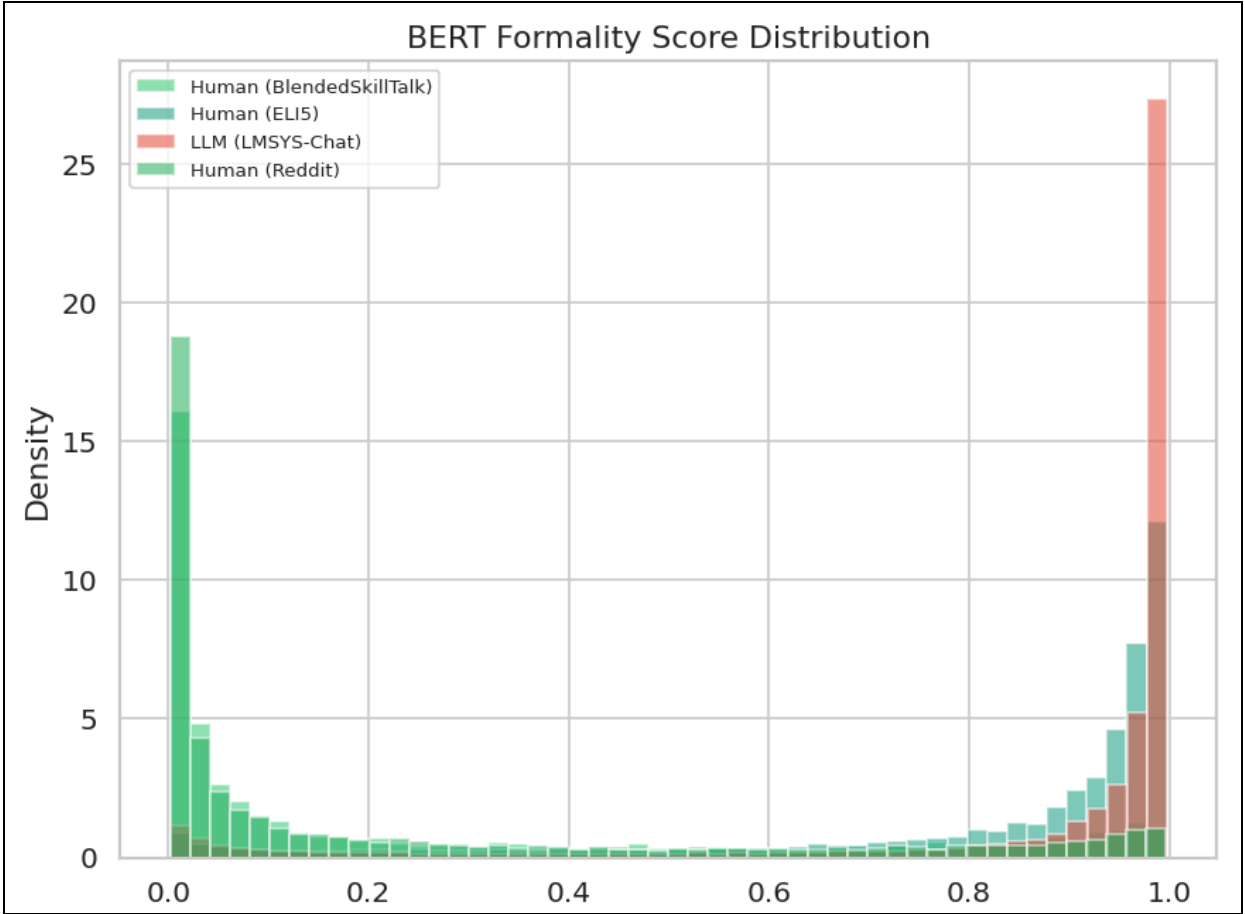
ROUGE-L	Estimates similarity or content preservation
Qualitative examples	Checks whether outputs remain coherent and useful

7. Results

7.1 LLMs Exhibit Higher Formality Than Human Baselines

The first result is that LMSYS LLM responses exhibit higher formality than the human baseline corpora. The main corpus-level results are shown below





Corpus	Mean FR	FR Standard Deviation	Mean RoBERTa P(formal)	RoBERTa Standard Deviation
BlendedSkillTalk	0.24	3.60	0.271	0.337
Reddit	0.52	4.07	0.245	0.325
ELI5	2.33	7.03	0.801	0.271
LLM / LMSYS-Chat	7.62	12.02	0.8585	0.271

The LLM corpus has the highest mean Formality Rate at 7.62 formal hits per 1,000 tokens. This is more than three times higher than ELI5, over fourteen times higher than Reddit, and over thirty times higher than BlendedSkillTalk. The high standard deviation of 12.02 shows that LLM formality is uneven: many responses contain few or no formal markers, while others contain a dense concentration of formal-register vocabulary.

The RoBERTa classifier supports the same broad finding. LMSYS responses have a mean RoBERTa P(formal) of 0.8585, which is higher than all three human baselines. ELI5 remains the closest baseline at 0.801, which is expected because ELI5 is explanatory rather than casual. Reddit and BlendedSkillTalk remain much lower, with means of 0.245 and 0.271, respectively. This result supports the claim that formal language bias is measurable through both explicit formal vocabulary and broader neural register classification.

7.2 Conversation Type Strongly Affects Formality

The conversation-type analysis shows that formality varies substantially by context.

Conversation Type	Formality Rate	RoBERTa P(formal)
Casual	1.9021	0.6326
General	8.0410	0.8525
Task	6.5871	0.8820

Task-oriented conversations have the highest RoBERTa P(formal), at 0.8820, which is expected because task responses often use structured explanations, instructions, and procedural language. General conversations have the highest Formality Rate, at 8.0410, suggesting that they contain the greatest concentration of explicit formal lexical markers. Casual conversations are lower on both metrics, with an FR of 1.9021 and RoBERTa P(formal) of 0.6326, but they still remain more formal than the most informal human baselines.

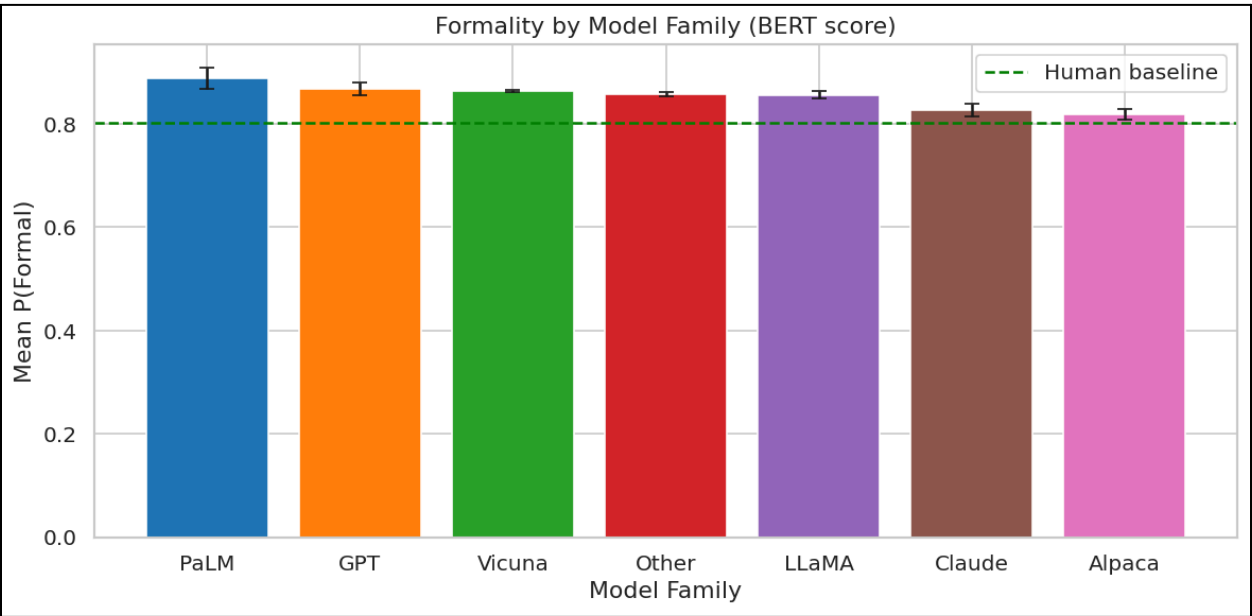
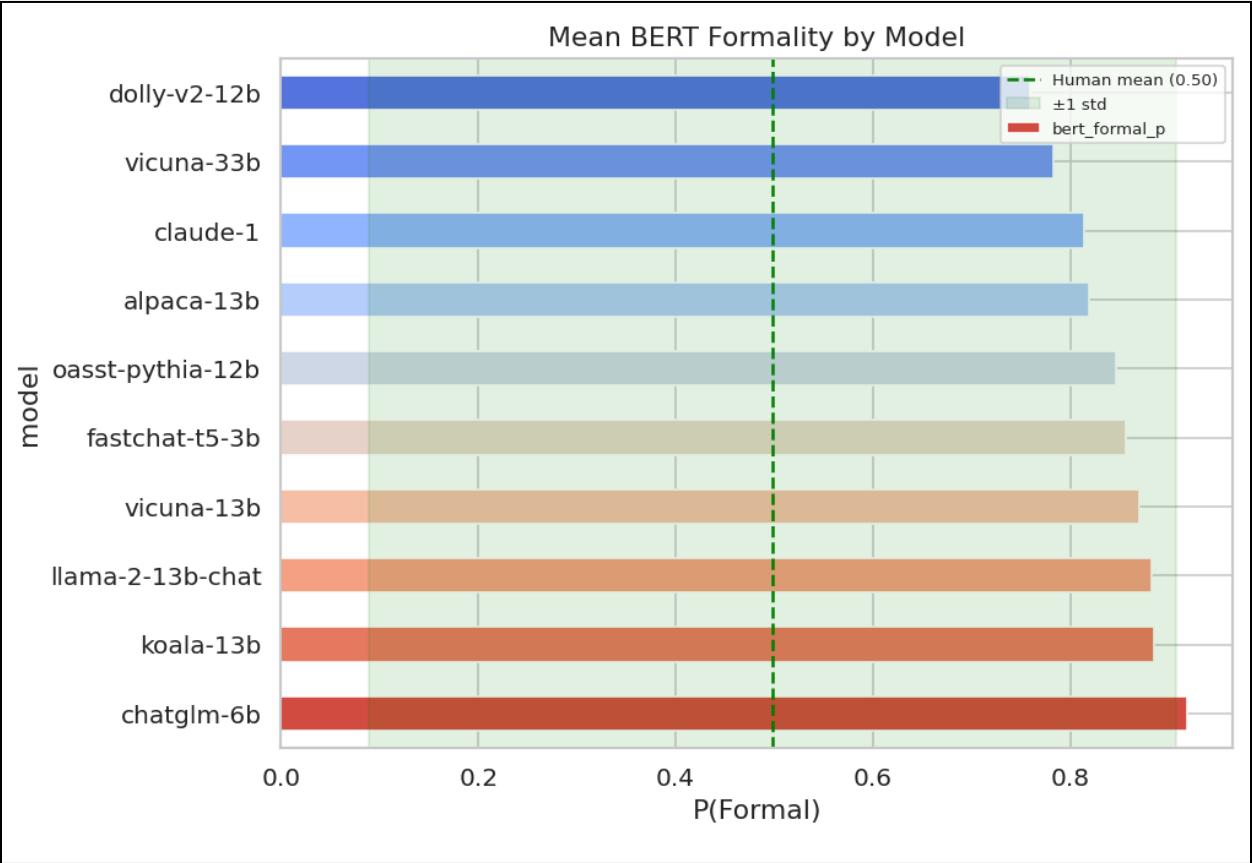
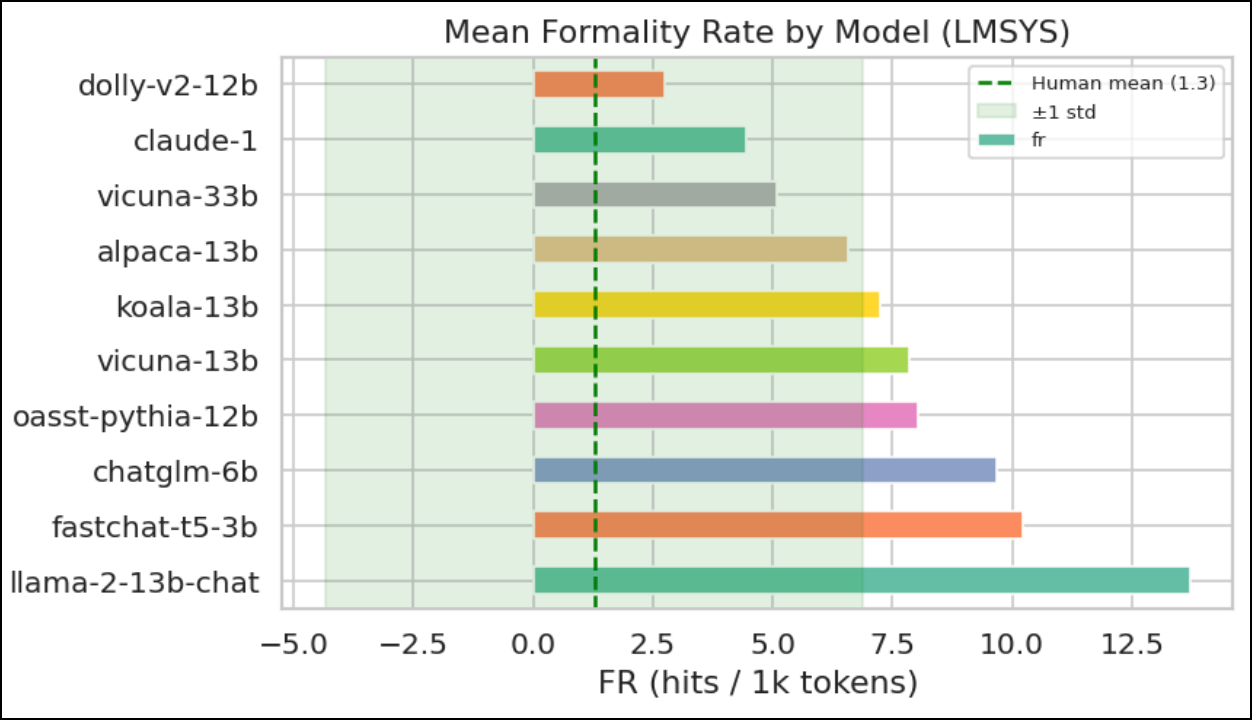
This is an important finding. It shows that formality bias is partly topic-dependent, but not entirely reducible to task type. Even when the context is casual, LLM outputs still show elevated formality compared with casual human dialogue.

7.3 Formality Bias Appears Across Model Families

The model-family analysis shows that formal language bias is widespread across LLM families. Nearly all evaluated families exceed human conversational baselines. However, the strength of the effect varies by family and by individual model.

The slide results indicate that PaLM and GPT show some of the strongest formal-register tendencies, while Claude appears lower but still near or above human baseline levels. This variation is important because it suggests that formality bias is not simply a universal property of transformer architecture. Instead, it likely reflects differences in pretraining data, instruction tuning, RLHF, and model-specific alignment strategies.

At the same time, the analysis cannot establish causality. Since model families differ across many uncontrolled dimensions, we cannot determine whether the bias comes specifically from pretraining data, supervised fine-tuning, RLHF, system prompting, or deployment environment. The result is best interpreted as correlational evidence that training and alignment pipelines influence formal register behavior.

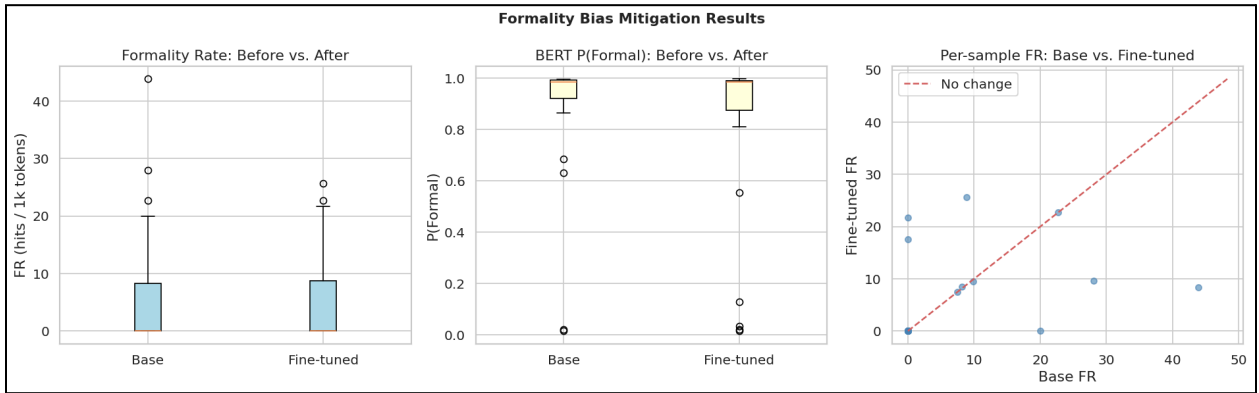


7.4 LoRA Fine-Tuning Reduces Formality

The LoRA mitigation experiment shows that lightweight adaptation can reduce formal register in Qwen2.5-3B-Instruct outputs. The base model’s mean Formality Rate decreases from 6.20 to 5.46 after fine-tuning, a 12.0% reduction in lexical formality. This suggests that the fine-tuned model uses fewer explicit formal-register words and phrases.

The RoBERTa P(formal) score also decreases, from 0.868 to 0.794. This is important because it suggests that the change is not only lexical. If only Formality Rate decreased, the model might simply be avoiding known formal words from the lexicon. Because the neural formality score also decreases, the results suggest a broader register shift.

ROUGE-L changes slightly from 0.0727 to 0.0766, so the fine-tuned outputs do not appear less similar to the informal references under this metric. However, ROUGE-L is a weak proxy for response quality, so this should be interpreted cautiously. Overall, the mitigation experiment supports the claim that register behavior is partially steerable, but it does not prove that lower formality always improves response quality.



Model	Mean FR	Mean RoBERTa P(formal)	ROUGE-L
Base Qwen2.5-3B-Instruct	6.2029	0.8681	0.0727
Fine-tuned LoRA	5.4601	0.7943	0.0766
Delta	-0.7429	-0.0738	+0.0039

8. Discussion and Analysis

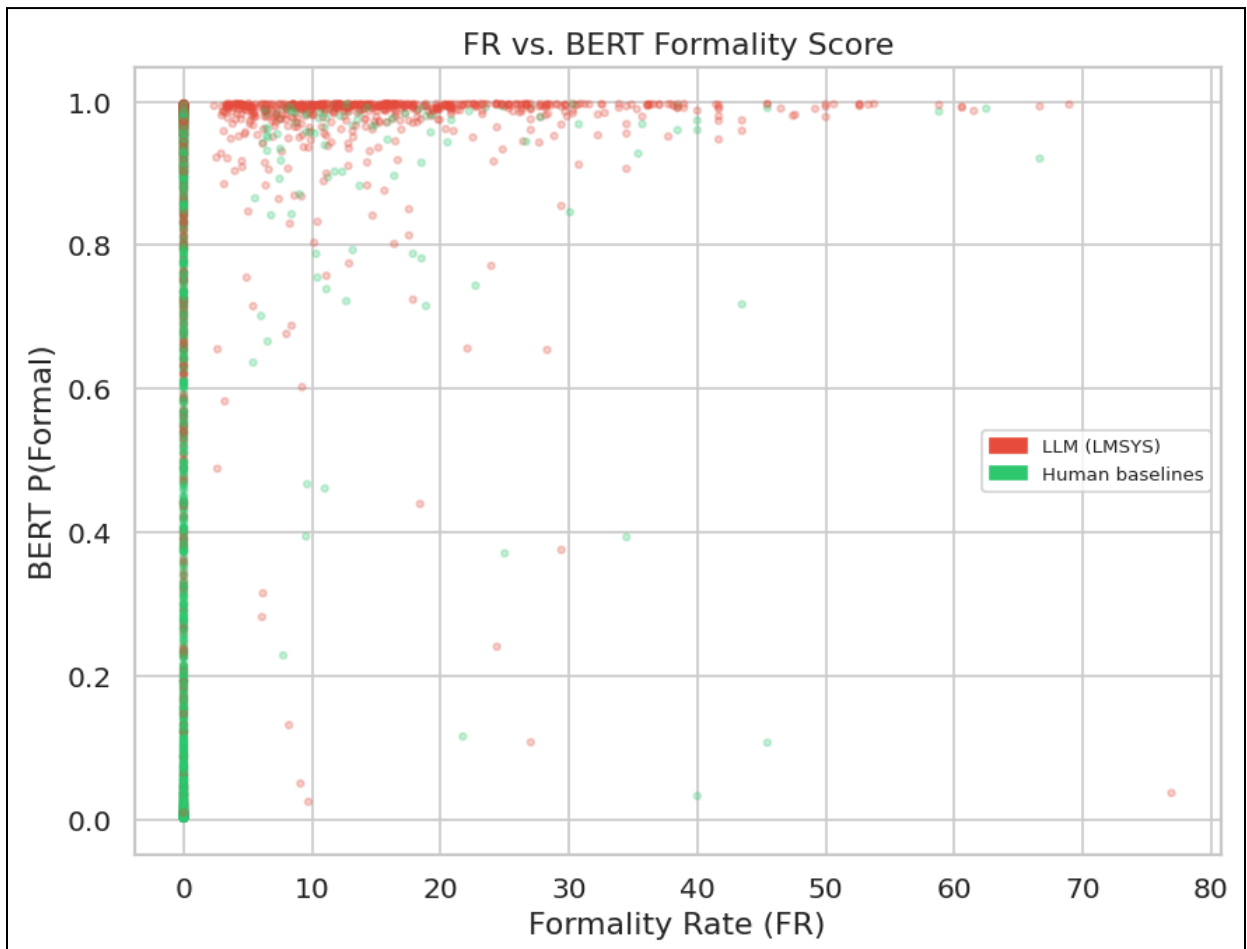
8.1 Formality Is Multidimensional

One of the clearest findings is that Formality Rate and RoBERTa P(formal) measure related but distinct aspects of register. Formality Rate measures direct lexical choices. It answers the question: how often does the response use formal words or phrases? RoBERTa P(formal), on

the other hand, captures broader stylistic cues such as sentence structure, explanatory tone, and discourse organization.

The difference between the metrics is especially visible in ELI5. ELI5 has a relatively low Formality Rate compared with LMSYS, at 2.33, but a high RoBERTa P(formal) of 0.801. This means ELI5 responses may not use many of the formal words in the lexicon, but their explanatory structure still appears formal to the classifier. This is not a contradiction. It shows that formality cannot be reduced to vocabulary alone.

This finding justifies the dual-metric framework. A lexicon-only approach would miss contextual or structural formality. A neural-only approach would be harder to interpret. Using both allows us to distinguish between explicit formal vocabulary and broader register classification.



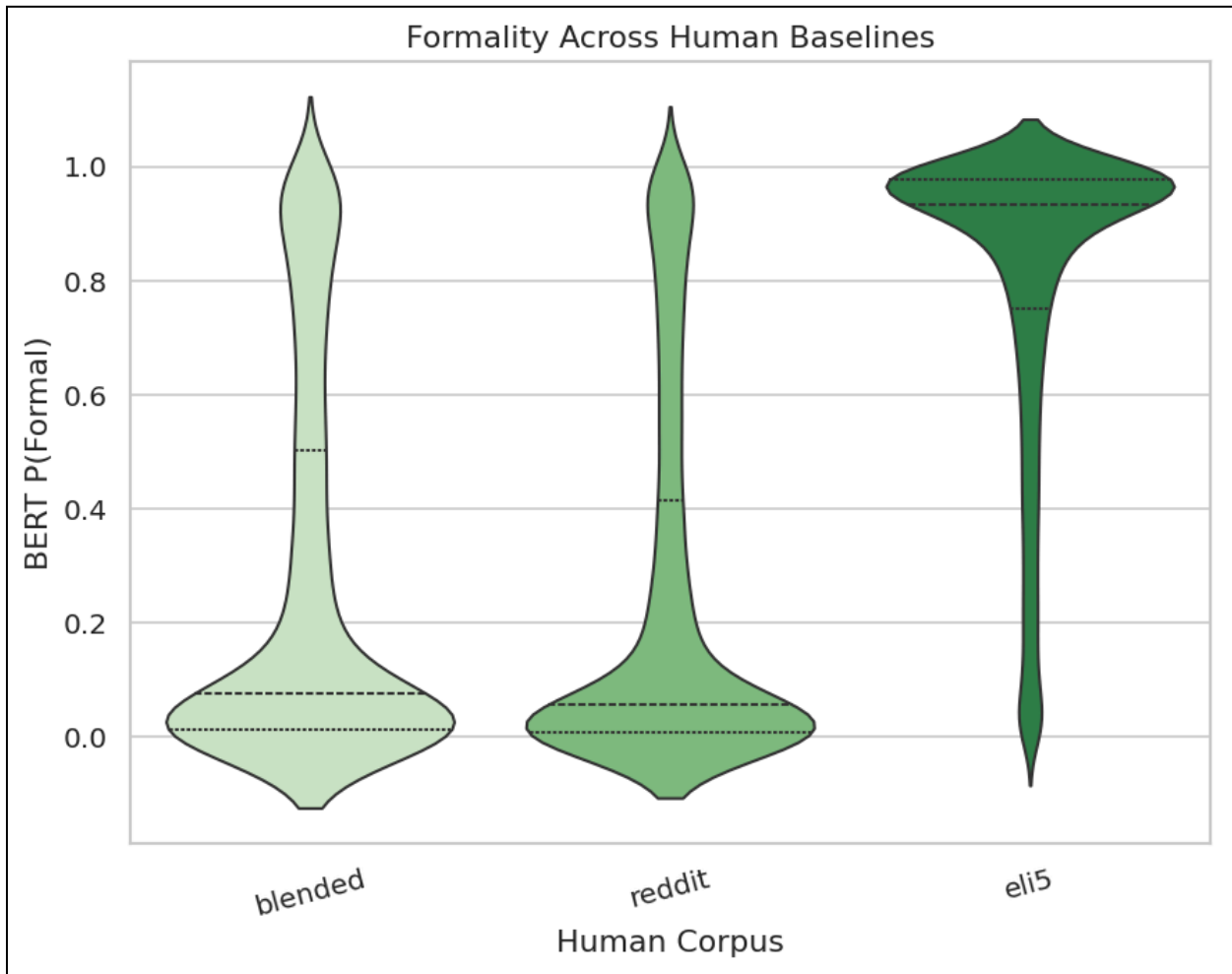
8.2 Baseline Choice Changes the Interpretation of Bias

Another major finding is that human baseline selection strongly affects the interpretation of LLM formality. Reddit and BlendedSkillTalk are highly informal, so LLMs appear dramatically more formal when compared against them. ELI5 is more formal because it contains explanatory answers, so the gap between LLMs and ELI5 is smaller.

This does not weaken the project’s conclusion. Instead, it makes the conclusion more precise. LLMs are not merely more formal than one arbitrarily chosen human dataset; they are

more formal than multiple baselines, including both casual dialogue and explanatory human writing. However, the size of the effect depends on the comparison corpus.

This means future work on LLM register bias should avoid treating “human text” as a single category. Human communication spans casual speech, educational explanation, technical writing, social media, academic prose, and many other genres. A good evaluation must compare LLM outputs to the appropriate human register for the task.



8.3 Formality Bias Is Topic-Dependent but Still Systematic

The conversation-type analysis shows that context strongly affects formality, but the two metrics emphasize different parts of that effect. This is expected. Task-oriented prompts naturally invite structured, explanatory, and sometimes formal responses, which helps explain why task responses have the highest RoBERTa P(formal). General responses, however, have the highest Formality Rate, meaning they contain the greatest density of explicit formal lexical markers. This difference reinforces that Formality Rate and RoBERTa P(formal) are related but not identical measures. For example, a request for technical help or academic explanation may require a more formal style than casual small talk.

However, the casual category remains more formal than casual human baselines. This suggests that LLMs may carry a default assistant register even when the user context does not require it. The model may be optimized to sound helpful, polished, safe, and complete. These

objectives can unintentionally produce language that resembles professional or academic writing rather than ordinary conversation.

Therefore, formality bias should not be understood as a fixed property of a model. It is better understood as an interaction between model training, prompt context, task type, and response conventions.

8.4 Model-Family Variation Suggests Training and Alignment Matter

The model-family results show that formal register bias is widespread but not identical across models. Some families show stronger formal tendencies than others. This suggests that training and alignment pipelines matter.

If formality bias were purely caused by transformer architecture, we would expect more uniform behavior across model families. Instead, differences across GPT, PaLM, Claude, LLaMA, Alpaca, and Vicuna suggest that pretraining data composition, instruction tuning, RLHF, and product-specific alignment all shape register behavior.

However, this analysis remains correlational. We cannot isolate individual causes without controlled comparisons between base models, instruction-tuned models, RLHF-tuned models, and models trained on different corpus mixtures. The current evidence supports the hypothesis that training matters, but it does not identify the exact causal source.

8.5 LoRA Shows Partial Register Control

The LoRA experiment suggests that register behavior can be shifted through lightweight fine-tuning, since both mean Formality Rate and mean RoBERTa P(formal) decrease after adaptation.

This result is useful as preliminary evidence because it suggests that formality may not be a fixed property of the base model. Instead, the decrease in mean scores suggests that register may be modifiable through targeted training, although this experiment does not establish a statistically significant mitigation effect. However, reducing formality is not always equivalent to improving response quality. In some settings, formal language is appropriate. The goal should not be to make all responses informal. A better long-term goal is register control: the ability to match the formality level to the user, task, and context.

The mitigation experiment also shows the limits of this evaluation. Although Qwen2.5-3B-Instruct is stronger than the earlier GPT-2-medium setup, the held-out prompt set is still small, and the paired statistical test did not show a significant reduction. Therefore, the

LoRA result should be interpreted as preliminary descriptive evidence of register steerability rather than definitive proof of mitigation.

9. Lessons Learned

9.1 Complementary Metrics Were Necessary

A major lesson from the project is that no single metric fully captures formality. Formality Rate is transparent and easy to inspect, but it only measures predefined lexical markers. RoBERTa captures broader style, but it is less interpretable and depends on the training distribution of the Pavlick formality scores dataset. The disagreement between FR and RoBERTa is not a failure; it reveals that register has multiple dimensions.

9.2 Baseline Selection Shaped Interpretation

The project also showed that baseline choice is central to register analysis. Comparing LLMs only to Reddit would make them appear extremely formal. Comparing only to ELI5 would make the effect appear smaller. Using several human baselines allowed us to separate casual conversational formality from explanatory human formality.

9.3 Register Mitigation Exposed Capacity Limits

The LoRA experiment showed that lightweight adaptation can influence register. However, it also highlighted the difficulty of changing style without affecting other aspects of generation. A model can become less formal while also becoming less coherent, less complete, or less precise. Future mitigation methods need to evaluate both register shift and response quality.

9.4 Data Preparation Influenced the Results

Preprocessing decisions had measurable effects on the analysis. Markdown artifacts, multilingual content, short responses, and formatting noise can all affect Formality Rate and RoBERTa scores. This reinforces the importance of careful data cleaning in stylistic analysis. Since the metrics operate directly on text, noisy text can easily become noisy evidence.

10. Limitations

10.1 Measurement Limitations

Both Formality Rate and RoBERTa P(formal) are proxy metrics. Formality Rate is limited by the lexicon. It can only count words and phrases that were included during lexicon construction. A response may sound formal because of syntax or tone even if it contains no formal lexicon hits. Conversely, a response might contain a formal word in an otherwise casual sentence.

RoBERTa is more context-sensitive, but it also has limitations. Because the classifier is trained on the Pavlick formality scores dataset, it may learn the formality distinctions specific to that dataset. Sentence-level formality data is not identical to multi-turn assistant conversations. Therefore, the classifier may not generalize perfectly to all conversational register styles.

10.2 Dataset Limitations

The datasets used in this project differ in genre, length, topic, and structure. Reddit is informal web text, BlendedSkillTalk is scripted dialogue, ELI5 is explanatory writing, and LMSYS contains model-generated assistant responses. These differences are useful for comparison, but they also introduce confounds.

Response length is another issue. Longer responses have more opportunities to contain formal markers. Although Formality Rate normalizes by token count, long structured responses may still differ systematically from short dialogue turns. Topic is also a confound because technical, academic, or explanatory topics naturally increase formality.

10.3 Causal Limitations

This project cannot definitively determine whether formality bias originates from pretraining, instruction tuning, RLHF, system prompting, or deployment conventions. Cross-family comparisons suggest that training and alignment matter, but they do not isolate causality.

A stronger causal design would compare controlled model variants: base models, instruction-tuned models, RLHF-tuned models, and models trained on different data mixtures. Without such controls, our conclusions about the source of bias must remain cautious.

10.4 Mitigation Limitations

The LoRA mitigation experiment is proof-of-concept rather than definitive debiasing. Although Qwen2.5-3B-Instruct is stronger than the earlier GPT-2-medium setup, the evaluation still uses

a small held-out prompt set. As a result, the observed reduction in mean formality does not fully establish generalization. The paired test did not show a statistically significant reduction, so the decrease in mean FR and mean RoBERTa P(formal) should be interpreted as descriptive rather than conclusive evidence of mitigation. Finally, lower formality is not always better. Some contexts require professional or technical language. The goal should be adaptive register control rather than universal informality.

11. Future Work

11.1 Scale Mitigation to Stronger Models

Future work should apply LoRA, DPO, or other lightweight adaptation methods to stronger models such as Mistral 7B or LLaMA-3-8B. This would test whether register mitigation scales to models with stronger instruction-following and language-generation capabilities.

11.2 Human-Centered Evaluation

Automatic metrics are useful but incomplete. Future work should include human evaluation of naturalness, trust, clarity, preference, and perceived helpfulness. Human raters could compare original and mitigated responses to determine whether lower formality actually improves user experience.

11.3 Causal Analysis of Bias Sources

A major next step is isolating the contributions of pretraining, supervised fine-tuning, RLHF, and task alignment. This would require comparing controlled model checkpoints or model families where training stages are known. Such analysis would help determine whether formality bias mostly comes from web-scale pretraining, instruction data, or human preference optimization.

11.4 Multi-Turn Register Drift

This project primarily evaluates individual responses. Future work should study how formality changes over multi-turn conversations. A model may become more formal over time, especially when giving explanations or maintaining assistant-like structure. Measuring register drift would better reflect real chatbot use.

11.5 Dynamic Lexicon Expansion

The current Formality Rate lexicon is manually curated. Future work could automatically expand the lexicon by contrasting LLM outputs against human corpora and identifying words or phrases that are disproportionately common in LLM responses. This would help detect emerging LLM-specific stylistic markers beyond manually selected examples.

12. Team Contributions

Team Member	Contributions
Oliver Krisetya	Designed and curated the two-tier formal-register lexicon, including Tier 1 LLM-specific markers and Tier 2 broad formal vocabulary. Implemented the Formality Rate computation pipeline, including tokenization, phrase matching, per-source aggregation, and visualization. Designed the conversation-type analysis and conducted model-family grouping. Ran statistical tests including Mann-Whitney U, Kruskal-Wallis, and paired tests. Diagnosed zero-inflation issues in the BlendedSkillTalk distribution and contributed to the interpretation of lexical formality results.
Warren Nguyen	Set up HuggingFace data collection and streaming pipelines for LMSYS-Chat-1M, BlendedSkillTalk, Reddit, and ELI5. Implemented markdown cleaning, response filtering, and dataset preprocessing. Fine-tuned the RoBERTa-base classifier on the Pavlick formality scores dataset and evaluated classifier performance using train/validation/test splits. Implemented the PEFT LoRA fine-tuning pipeline using TRL SFTTrainer on Qwen2.5-3B-Instruct. Constructed the low-formality supervised fine-tuning dataset, computed ROUGE-L evaluation, performed qualitative error analysis on held-out prompt pairs, exported final results, and organized the final notebook.

13. Conclusion

This project investigated whether large language models exhibit measurable formal language bias. Using a dual metric framework, we found that LLM outputs from LMSYS-Chat-1M are more formal than multiple human baselines according to both lexicon-based Formality Rate and

RoBERTa-based formal probability. The effect varies by conversation type and model family, suggesting that formality bias is shaped by task context, training data, and alignment procedures.

The project also showed preliminary evidence that mitigation is possible. LoRA fine-tuning on low-formality data produced lower mean Formality Rate and mean RoBERTa P(formal) in Qwen2.5-3B-Instruct outputs, suggesting that register may be partially steerable through lightweight adaptation. However, because the held-out evaluation set was small and the paired test was not statistically significant, the mitigation results should be treated as preliminary rather than definitive. However, the findings also reveal important limitations. Formality is multidimensional, automatic metrics are proxies, human baselines are genre-dependent, and causal sources of bias remain difficult to isolate.

Overall, the project supports the conclusion that LLM formality bias is not merely anecdotal. It is measurable, widespread, and potentially steerable, though the mitigation result remains preliminary. Future work should focus on stronger models, human-centered evaluation, causal analysis, and adaptive register control rather than simply reducing formality in all contexts.